

IT Security 2

Unit 10 - Blocking Inappropriate Web Traffic with Advanced URL Filtering

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Announcements

Blocking Inappropriate Web Traffic with Advanced URL Filtering

Some slides were adapted from Palo-Alto for use in this class. Content is a mesh between vendor provided information and instructor comments.

BLOCKING INAPPROPRIATE WEB TRAFFIC WITH ADVANCED URL FILTERING



AVOID DANGEROUS NETWORK NEIGHBORHOODS

- Advanced URL Filtering Overview
- Configuring Advanced URL Filtering Security Profiles
- Attaching Advanced URL Filtering Profiles to Policy Rules

Learning Objectives

After you complete this module, you should be able to:

- Configure the firewall to block traffic from known-malicious IP addresses
- Configure the firewall to block traffic from known-malicious domains
- Configure the firewall to block traffic from known-malicious URLs
- Describe other Advanced URL filtering operations and options



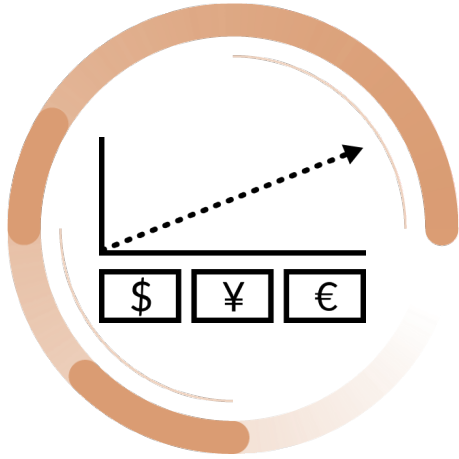


Advanced URL Filtering Overview

Configuring Advanced URL Filtering Security Profiles

Attaching Advanced URL Filtering Profiles to Policy Rules

Challenges with Preventing Web-Based Threats



Enable business without compromising security

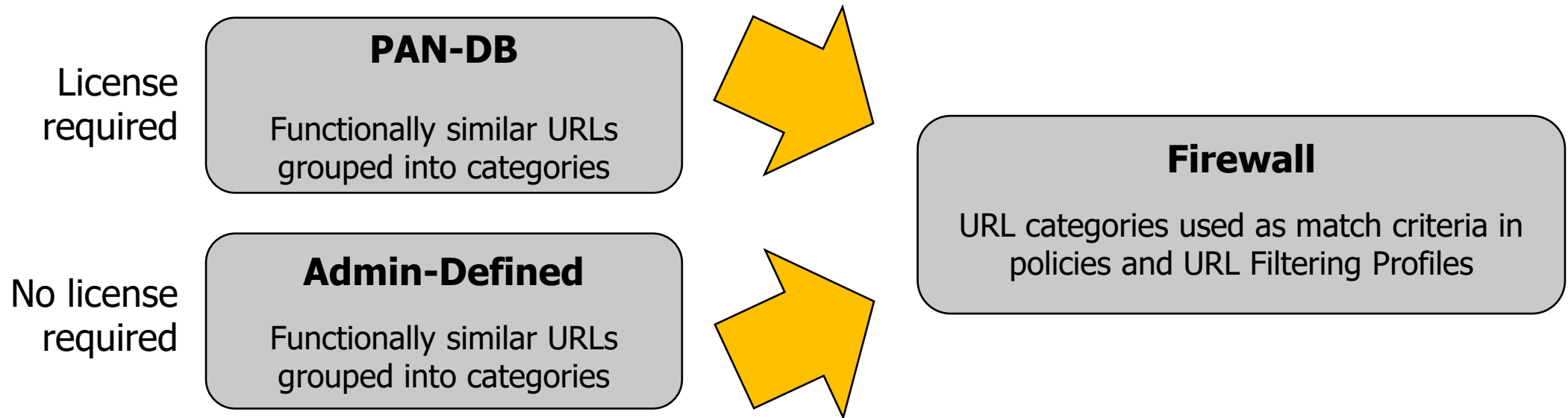


Encrypted web content



Silo management

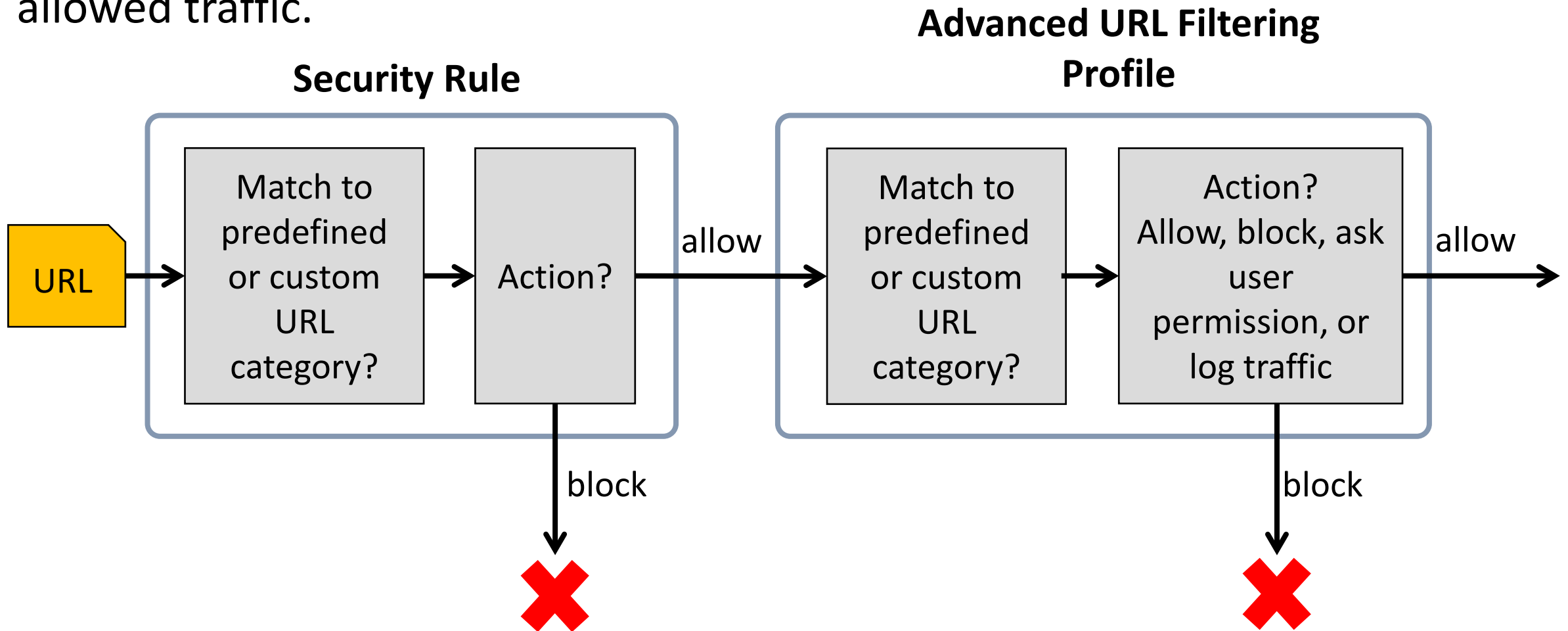
Advanced URL Filtering Features



- Use URL filtering to reduce the attack surface:
 - Disrupts the Delivery or Command-and-Control stage of the cyberattack lifecycle
- Two methods:
 - Use URL categories as a match condition in a Security policy deny rule.
 - Block access to a URL category in an Advanced URL Filtering Profile.

Advanced URL Filtering Profiles

Advanced URL Filtering Profiles implement additional security checks on allowed traffic.



URL Category: Policy Versus Profile

Policies > Security

	NAME	TYPE	Source			Destination		APPLICATION	SERVICE	URL CATEGORY	ACTION	PROFILE
			ZONE	ADDRESS	USER	ZONE	ADDRESS					
1	Block-Social-Media-Access	universal	 Users_Net	any	any	 Internet	any	any	any	social-networking	 Deny	none
2	Allow-Web-Mail	universal	 Users_Net	any	any	 Internet	any	any	any	web-based-email	 Allow	

URL Category in a Policy

Used as a match condition

URLs matched to predefined or custom URL categories

Action determined in the policy rule

URL category name logged in the URL Filtering log

Advanced URL Filtering Security Profile

Applied to traffic allowed by Security policy

URLs matched to predefined or custom URL categories

Action more granularly configured for individual URLs or URL categories

URL details logged in the URL Filtering log

URL Filtering Log

- Attachment of a URL Filtering Profile to a Security rule generates log entries:
 - “alert,” “block,” “continue,” and “override” actions trigger log entries.

Monitor > Logs > URL Filtering

	RECEIVE TIME	CATEGORY	URL CATEGORY LIST	URL	FROM ZONE	TO ZONE	SOURCE	DESTINATION	APPLICATION	ACTION
	03/23 19:24:22	proxy-avoidance-and-anonymizers	proxy-avoidance-and-anonymizers,low-risk	hidester.com/fav...	Users_Net	Internet	192.168.1.20	104.26.13.83	web-browsing	block-url
	03/23 19:24:22	proxy-avoidance-and-anonymizers	proxy-avoidance-and-anonymizers,low-risk	hidester.com/log...	Users_Net	Internet	192.168.1.20	104.26.13.83	web-browsing	block-url
	03/23 19:24:22	proxy-avoidance-and-anonymizers	proxy-avoidance-and-anonymizers,low-risk	hidester.com/	Users_Net	Internet	192.168.1.20	104.26.13.83	web-browsing	block-url
	03/23 19:24:22	proxy-avoidance-and-anonymizers	proxy-avoidance-and-anonymizers,low-risk	hidester.com/	Users_Net	Internet	192.168.1.20	104.26.13.83	web-browsing	block-url
	03/23 19:20:47	hacking	hacking,low-risk	hacker9.com/fav...	Users_Net	Internet	192.168.1.20	159.89.148.144	web-browsing	block-url
	03/23 19:20:47	hacking	hacking,low-risk	hacker9.com/log...	Users_Net	Internet	192.168.1.20	159.89.148.144	web-browsing	block-url
	03/23 19:20:47	hacking	hacking,low-risk	hacker9.com/	Users_Net	Internet	192.168.1.20	159.89.148.144	web-browsing	block-url
	03/23 19:20:47	hacking	hacking,low-risk	hacker9.com/	Users_Net	Internet	192.168.1.20	159.89.148.144	web-browsing	block-url

Advanced URL Filtering Overview



Configuring Advanced URL Filtering Security Profiles

Attaching Advanced URL Filtering Profiles to Policy Rules

URL Filtering Security Profile

Objects > Security Profiles > URL Filtering

☐	NAME ▾	SITE ACCESS	USER CREDENTIAL SUBMISSION	HTTP HEADER INSERTION	DESCRIPTION
☐	default	Allow Categories (58) Alert Categories (5) Continue Categories (0) Block Categories (10) Override Categories (0)	Allow Categories (73) Alert Categories (0) Continue Categories (0) Block Categories (0)		
☐		Allow Categories (61) Alert Categories (3) Continue Categories (0) Block Categories (9) Override Categories (0)	Allow Alert C Contin Block C		Default URL-Filtering profile for security rules.

Predefined profile

Click each item to display categories in the list.

+ Add - Delete ↺ Clone 📄 PDF/CSV (* indicates custom URL category, + indicates external dynamic list)

- To create customized profiles:
 - Clone the default read-only profile and edit the clone, or
 - Add a brand-new profile

URL Filtering Security Default Categories

URL Filtering Profile (Read Only) ?

Name

Description

Categories | URL Filtering Settings | User Credential Detection | HTTP Header Insertion | Inline Categorization

73 items → X

☐ CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
Pre-defined Categories		
☐ abortion	allow	allow
☐ abused-drugs	block	block
☐ adult	block	block
☐ alcohol-and-tobacco	allow	allow
☐ auctions	allow	allow
☐ business-and-economy	allow	allow
☐ command-and-control	block	block

* indicates a custom URL category, + indicates a custom URL category

[Check URL Category](#)

Pre-defined URL categories

Malicious URLs are configured as *block* in the default profile

Multi-Category and Risk-Based URL Filtering

Objects > Custom Objects > URL Category

Custom URL Category

Name

Marketing-Department

Description

Custom URL Filter for the Marketing Department

Type

Category Match

Matches all of the following categories

Q

☐

CATEGORIES

☐

low-risk

☐

medium-risk

☐

high-risk

☐

newly-registered-domain

+

Add

-

Delete

- PAN-DB Advanced URL Filtering assigns websites to multiple categories.
- Categories indicate:
 - The site's risk
 - The site's content
 - The site's purpose or function
- The security-related risk categories demonstrate levels of suspicious activity.
- Websites registered for fewer than 32 days are assigned to *new-registered-domain*.

For a Tech Doc about this topic, log into Live and search for "Multi-Category URL Filtering"

Configure Per-URL Category Actions

URL Filtering Profile (Read Only)

Name: default

Description:

Categories | URL Filtering Settings | User Credential Detection | HTTP Header Insertion | Inline Categorization

73 items

CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
Pre-defined Categories		
☐ abortion	allow	allow
☐ abused-drugs	block	block
☐ adult	block	block
☐ alcohol-and-tobacco	allow	allow
☐ auctions	allow	allow
☐ business-and-economy	allow	allow
☐ ...	block	block

* indicates a custom URL category

Check URL Category

Has drop-down list with option to change all actions

Admin-defined URL categories – replaces overrides

Action to take when URL is accessed; "allow" is default

alert
allow
block
continue
override
none

alert
allow
block
continue
none

Action to take if user submits credentials to allowed URL

URL matching order:

1. Custom URL categories*
2. External Dynamic Lists*
3. PAN-DB firewall cache
4. PAN-DB cloud

*Supports wildcard characters

Configure a Custom URL Category

Configure URL filtering based on specific URLs or URL categories.

Objects > Custom Objects > URL Category > Add

Custom URL Category

Name: Match-Shopping-URL-List

Description: Shopping URL match list

Type: URL List

Matches any of the following URLs, domains or host names

☐	SITES
☐	*.newegg.com
☐	*.amazon.com
☐	*.pricegrabber.com

Three specific "shopping" websites

+ Add - Delete | Import Export

Enter one entry per row.
Each entry may be of the form www.example.com or it could have wildcards like www.*.com.

Custom URL Category

Name: Match-URL-Categories

Description: Category URL match list

Type: Category Match

Matches all of the following categories

☐	CATEGORIES
☐	shopping
☐	auctions
☐	games

All "shopping" websites plus two other categories

+ Add - Delete

URL Filtering Response Pages



Web Page Blocked

The web page you are trying to visit has been blocked in accordance with company policy. Please contact your system administrator if you believe this is an error.

User: 192.168.1.20

URL: www.betus.com.pa/?token=Z-DI0obEUqaxonBEqMydTmNd7ZgqdRLk

Category: gambling



Web Page Blocked

The web page you are trying to visit has been blocked in accordance with company policy. Please contact your system administrator if you believe this is an error.

User: 192.168.1.20

URL: http://hacker9.com/

Category: hacking

If you click Continue to proceed to the page, this action will be logged.

[Continue](#)

[Return to previous page](#)



Web Page Blocked

The web page you are trying to visit has been blocked in accordance with company policy. Please contact your system administrator if you believe this is an error.

User: 192.168.1.20

URL: http://hidester.com/

Category: proxy-avoidance-and-anonymizers

If you require access to this page, have an administrator enter the override password here:

[Continue](#)

[Return to previous page](#)

URL Admin Settings

Device > Setup > Content-ID > URL Admin Override > Add

URL Admin Override

?

Password

.....

Confirm Password

.....

SSL/TLS Service Profile

None

▼

Mode

☐ Transparent

☒ Redirect

Address

10.30.11.51

- Configure URL Admin Override password.

Device > Setup > Content-ID > URL Filtering

URL Filtering

?

URL Continue Timeout (min)

15

URL Admin Override Timeout (min)

15

URL Admin Lockout Timeout (min)

30



☐ Hold client request for category lookup

Category lookup timeout (sec)

2

PAN-DB Server

Cloud list separated by commas

- Configure URL Admin Override password timeout period.

Configure Safe Search and Logging Options

Objects > Security Profiles > URL Filtering > Add

URL Filtering Profile

Name

External-URL-Filtering

Description

Standard corporate URL profile for all security policy rules.

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

☒ Log container page only

☒ Safe Search Enforcement

HTTP Header Logging

☐ User-Agent

☐ Referrer

☐ X-Forwarded-For

Has dedicated block page. See **Device > Response Pages.**

HTTP Header Insertion and Modification

- Limit access to only enterprise versions of SaaS applications.
- Four predefined SaaS applications:
 - Dropbox
 - Google
 - Office 365
 - YouTube
- Inserts HTTP header if missing or overwrites existing header.
- **Dynamic Fields** inserts *X-Authenticated-User* header to specify user's name and domain to secondary devices:
 - To enforce additional user-based policy controls

Objects > Security Profiles > URL Filtering > Add

HTTP Header Insertion

Name: Outbound-GoogleURL-HTTP-Header

Type: Google Apps Access Control

Domains:

- *.google.com
- gmail.com

For traffic to google.com or gmail.com, add the header and user's domain name.

Headers:

☐	HEADER	VALUE	LOG
☐	X-GooGApps-Allowed-Domains		☐

+ Add - Delete

Inline Categorization

Objects > Security Profiles > URL Filtering

URL Filtering Profile

Name

External-URL-Filtering

Description

Standard corporate URL profile for all security policy rules

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

☒ Enable local inline categorization

☒ Enable cloud inline categorization

Exceptions

☐ CUSTOM URL CATEGORY/EDL ^

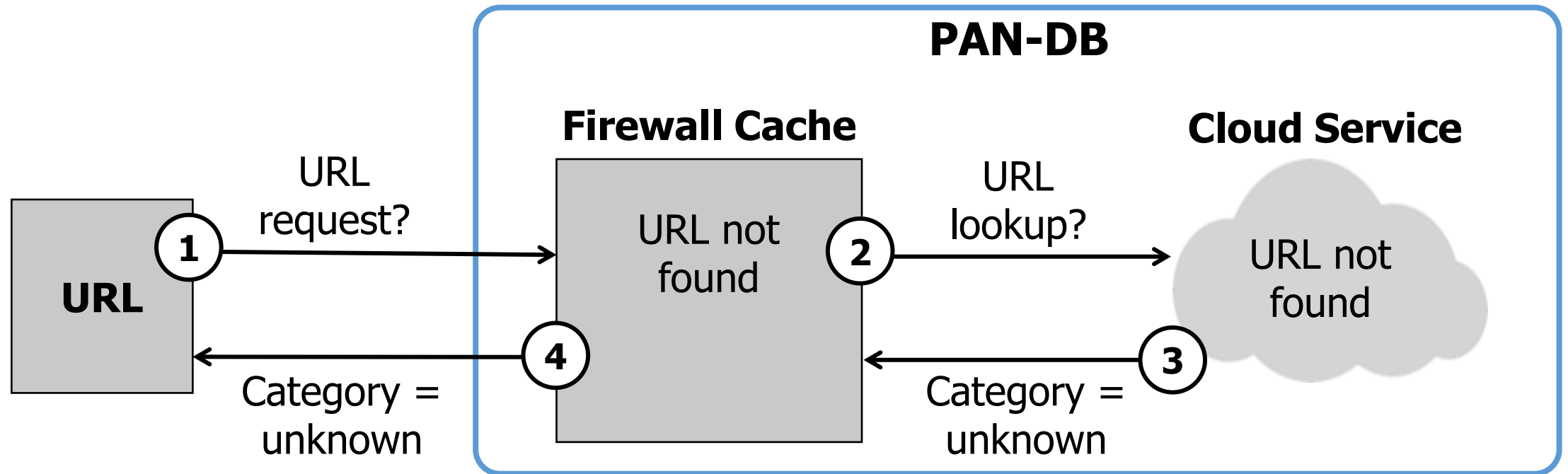
+ Add

- Delete

(Optionally) Add URL exceptions to exclude specific URLs.

Recommendations for Unknown URL Category

Category column in URL Filtering log lists *unknown*.

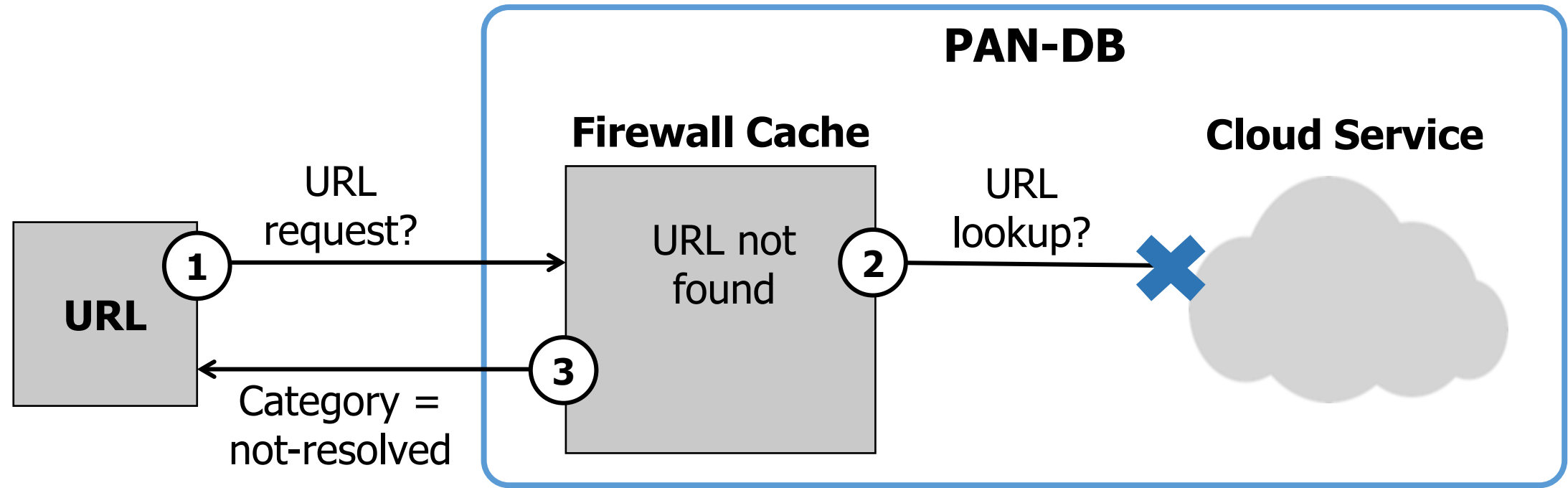


Recommendation: Set unknown URL category action to support your security requirements.

For a Tech Doc about this topic, log into Live and search for "Malicious URL Categories"

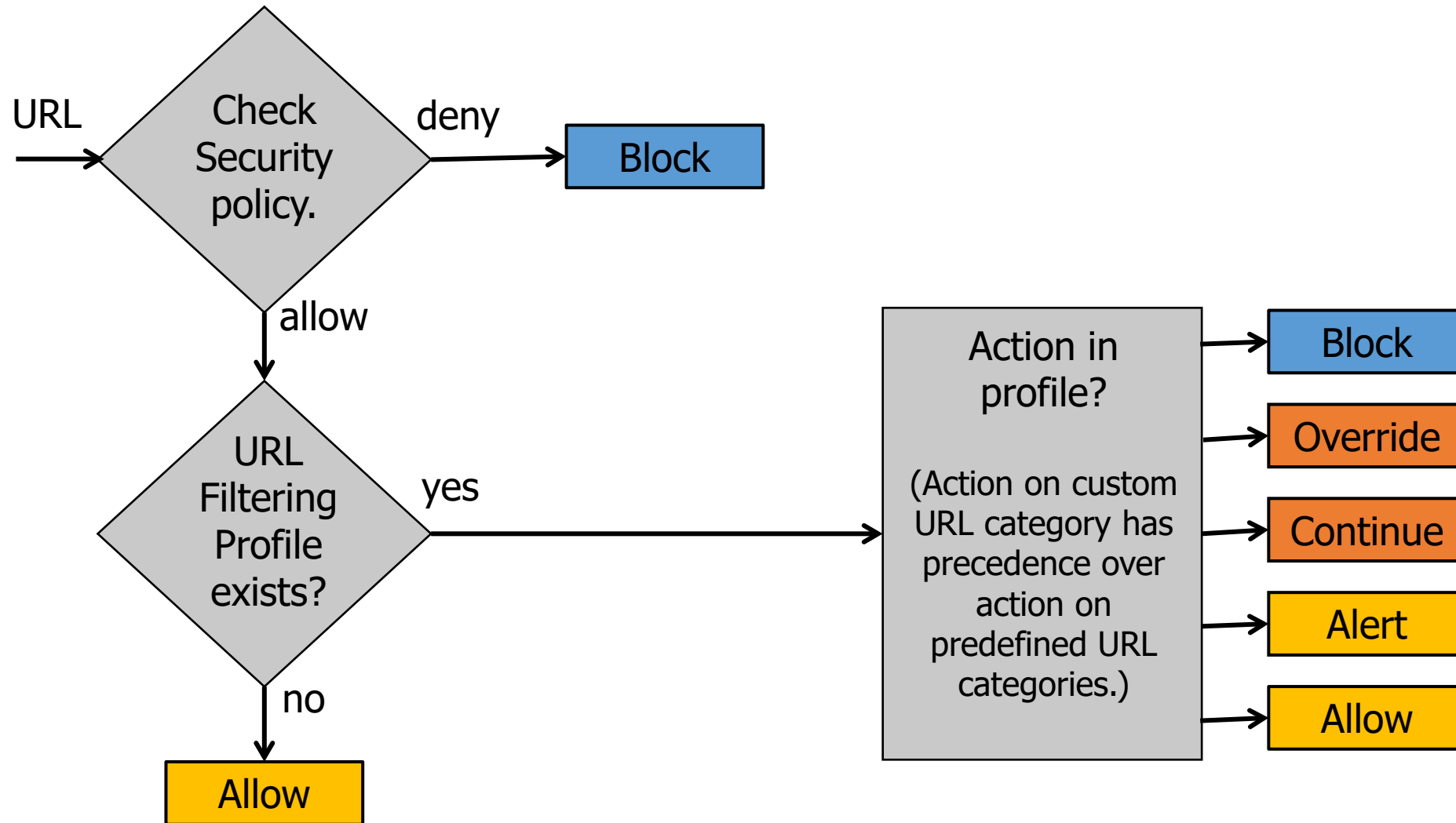
Recommendations for Not-Resolved URL Category

Category column in URL Filtering log lists *not-resolved*.



Recommendation: Set *not-resolved* URL category match action to “alert.”

Advanced URL Filtering Action Precedence



Advanced URL Filtering Precedence Example

Allow **Pentesters-Grp** to access all **hacking** category URLs except **www.hackers9.com**.

Policies > Security

	NAME	TAGS	TYPE	Source			Destination		APPLICATI...	SERVICE	URL CATEGORY	ACTION	PROFILE	OPTIONS
				ZONE	ADDRESS	USER	ZONE	ADDRESS						
1	Allow-Pentesters	Users_Ne...	universal	Users_Net	any	Pentesters-Grp	Internet	any	any	application-default	any	Allow		

Outbound-URL-Filt
profile

1

Objects > Custom Objects > URL Category

Custom URL Category

Name: Denied-Hacking-URLs

Description: URL Category to Block Hacking URLs

Type: URL List

Matches any of the following URLs, domains or host names

1 item → ×

SITES
☐ www.hackers9.com

URL Filtering Profile

Name: Corporate-URL-Filtering

Description: Default URL-Filtering profile for security rules.

Categories | URL Filtering Settings | User Credential Detection | HTTP Header Insertion | Inline Categorization

CATEGORY	SITE ACCESS
Custom URL Categories	
☒ Denied-Hacking-URLs *	2 block
Pre-defined Categories	
☐ hacking	3 continue

Action
taken

2

3

Recategorization Request: Via Log Entries

Monitor > Logs > URL Filtering

	RECEIVE TIME	CATEGORY	URL CATEGORY LIST	URL
	07/17 19:45:27	News-Sites	News-Sites,news,low-risk	a57.foxnews.com/
	07/17 19:45:27	News-Sites	News-Sites,news,low-risk	static.foxnews.com/
	07/17 19:45:27			

Details

Severity informational

Repeat Count 1

URL static.foxnews.com/

HTTP Method

[Request Categorization Change](#)

Request Categorization Change

URL

Log Category

Current Category

Categorization on the server has been updated since this log entry was generated

Suggested Category

[get descriptions](#)

Email

Confirm Email

Comments

The following characters are not supported: ";|' & '\.

For a Tech Doc about this topic, log into Live and search for "Request to Change the Category for a URL"

Recategorization Requests: Via Webpage

Objects > Security Profiles > URL Filtering > Add

Custom URL Categories

- ☐ Denied-Hacking-URLs *
- ☐ Risky-URLs *
- ☐ URLs-Block-Per-Company-Policy *

Pre-defined Categories

- ☐ abortion
- ☐ abused-drugs

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

Test A Site

URL

Which URL would you like to query once more?

SEARCH

URL: www.raptitude.com

Category: Personal Sites and Blogs

Description: Personal websites and blogs by individuals or groups.

Example Sites: www.blogspot.com, www.wordpress.com, www.greatamericanphotocontest.com

Category: Low Risk

Description: Sites that are not medium or high risk are considered low risk. These sites have displayed benign activity for a minimum of 90 days. The low risk category includes both sites that have a history of only benign activity, and sites found to be malicious in the past, but that have displayed benign activity for at least 90 days.

Example Sites: www.google.com, www.schwab.com, www.amazon.com

[Request Change](#)

Palo Alto Networks URL fil x

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paloalto NETWORKS

Network Security Secure Access Service Edge Cloud-Native Security Security Operations

Advanced URL Filtering Overview

Configuring Advanced URL Filtering Security Profiles

Attaching Advanced URL Filtering Profiles to Policy Rules



Use a URL Filtering Profile

- Attach the URL Filtering Profile to a Security policy allow rule.

Policies > Security

	NAME	TAGS	TYPE	Source			Destination		APPLICATI...	SERVICE	URL CATEGORY	ACTION	PROFILE	OPTIONS
				ZONE	ADDRESS	US...	ZONE	ADDRESS						
1	Allow-Some-Web-Access	Users_Net	universal	Users_Net	any	any	Internet	any	any	application-default	any	Allow		

- Profile can individually block, allow, or log access to URL categories:
 - Block: command-and-control, dynamic-dns, hacking, high-risk, malware, phishing, unknown.
 - Consider blocking: adult, extremism, new-registered-domain, parked, proxy-avoidance-and-anonymizers, questionable.
- Enable logging for better visibility in logs and reports.

Assigning URL Profile to Security Rules

Policies > Security > Add

Security Policy Rule

General | Source | Destination | Application | Service/URL Category | **Actions** | Usage

Action Setting

Action: Allow

☐ Send ICMP Unreachable

Profile Setting

Profile Type: Profiles

Antivirus: Corp-AV

Vulnerability Protection: Corp-Vuln

Anti-Spyware: Corp-AS

URL Filtering: Corp-URL-Filter

File Blocking: Corp-FileBlock

Data Filtering: Corp-DataFilter

WildFire Analysis: None

Profile Setting Modal

Profile Type: Group

Group Profile: Corp-Profiles-Group

- Assign individual Security Profiles to a Security policy rule.
- or
- Assign a Security Profile Group to a Security policy rule.

Module Summary

Now that you have completed this module, you should be able to:

- Configure the firewall to block traffic from known-malicious IP addresses
- Configure the firewall to block traffic from known-malicious domains
- Configure the firewall to block traffic from known-malicious URLs
- Describe other Advanced URL Filtering operations and options



Questions

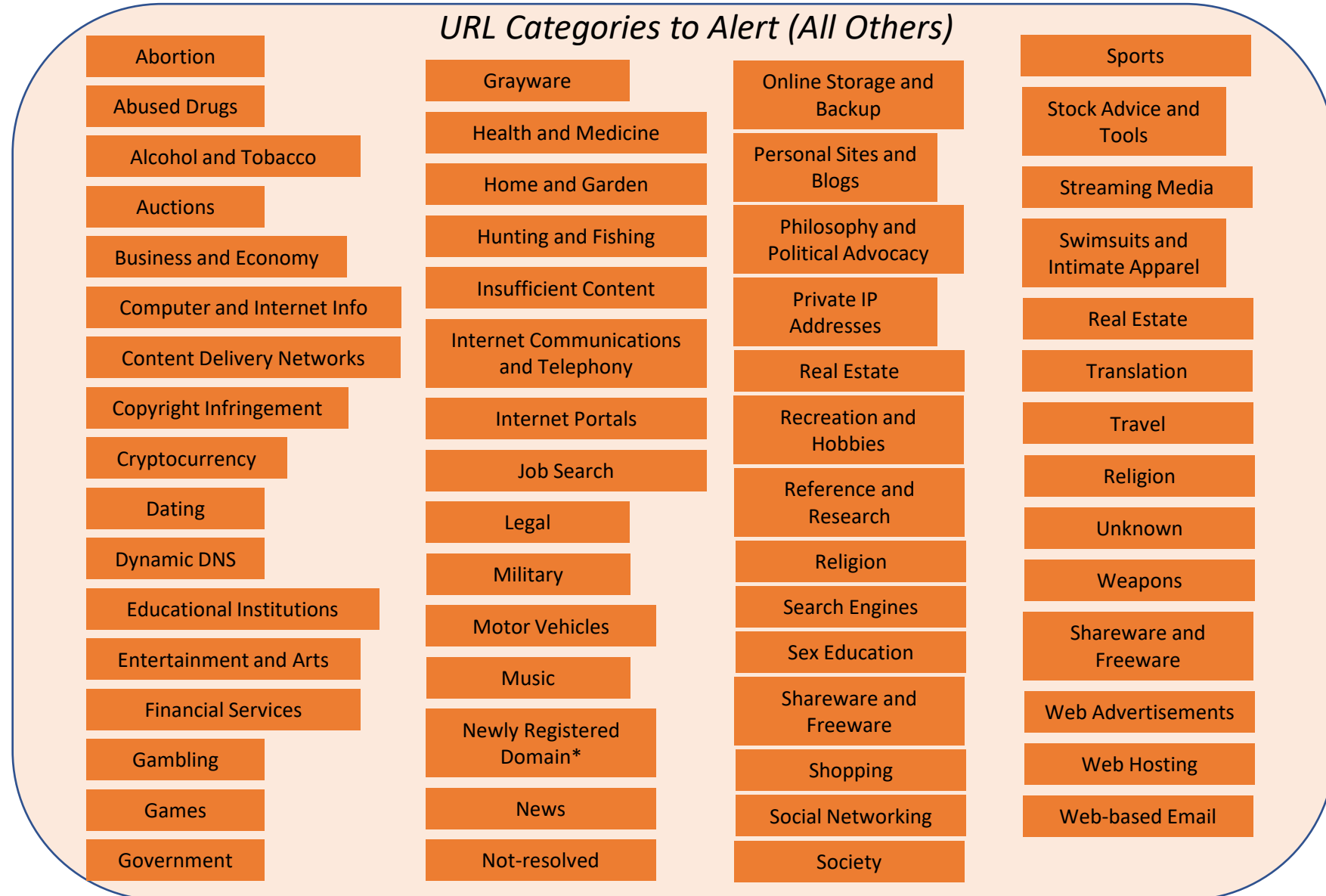


Lab 9 Overview

URL Categories to Block



URL Categories to Alert (All Others)



Lab 9: Blocking Inappropriate Web Traffic with Advanced URL Filtering

- Test access to inappropriate web content without URL blocking in place
- Create a Security Policy rule to block inappropriate web content using the URL Category
- Test the Security Policy rule and examine the results
- Disable the Security Policy rule
- Create and apply a URL Filtering Profile to block access to a malicious URL
- Test the Security Profile and examine the results

End Content

Unit 9 Homework

- Always check Blackboard for assignments and when they are due.
- Unit 10 Lab: Netlabs Lab 9 – Blocking Inappropriate Web Traffic with Advanced URL Filtering
- Unit 9 Quiz